Backend Project Suggestions

Great choice. This is a **very strong backend project** and will look excellent on your resume. We'll build it in a **structured way like a real production system**, not just a small college project.

I'll guide you step-by-step.

AI-Powered Personal Finance Tracker – Project Plan

Final Features

By the end, your project will have:

Core Features

- User Signup / Login (Authentication)
- Add income / expense transactions
- Expense categories (food, travel, bills, etc.)
- Monthly analytics dashboard
- Transaction history

AI Features

- AI spending insights
- AI budget suggestions
- AI monthly summary

Example output:

> "You spent 42% of your income on food this month. Consider reducing restaurant expenses."

Tech Stack

Backend

- **Flask**
- **Flask-Login**
- **SQLAlchemy**

- **REST APIs**

Database

- **MySQL**

Frontend

- HTML
- Bootstrap
- Jinja templates

AI

- **OpenAI API** *(or rule-based AI if API not used)*

Tools

- VS Code
- GitHub

System Architecture

```
Frontend (HTML + Bootstrap)
        |
        |
Flask Backend (REST APIs)
        |
Business Logic
        |
SQLAlchemy ORM
        |
MySQL Database
        |
AI Insight Engine
```

Database Design

We will create **3 main tables**

Users

id	username	email	password_hash

Transactions

id	user_id	amount	category	type	date

type =

- income
- expense

Categories

| id | name |

Examples

- Food
- Travel
- Rent
- Shopping

Project Folder Structure

```
finance-tracker/
│
├── app.py
├── config.py
├── models.py
├── requirements.txt
│
├── routes/
│   ├── auth_routes.py
│   ├── transaction_routes.py
│   └── insights_routes.py
│
├── templates/
│   ├── base.html
│   ├── login.html
│   ├── register.html
│   ├── dashboard.html
│   └── add_transaction.html
│
├── static/
```

```
│    ├──── css
│    └──── js
│
└──── ai/
    └──── insight_engine.py
```

This structure itself impresses interviewers.

Development Roadmap

We will build in **7 steps**

Step 1

Environment setup

Step 2

User authentication

Step 3

Database models

Step 4

Add transactions

Step 5

Dashboard analytics

Step 6

AI insight engine

Step 7

Deployment

Step 1 — Environment Setup

Open terminal.

Create project folder

```bash
mkdir finance-tracker
cd finance-tracker
```

Create virtual environment

Windows:

```bash
python -m venv venv
venv\Scripts\activate
```

Install dependencies

```bash
pip install flask flask-login flask-sqlalchemy pymysql openai pandas
```

Create requirements file

```bash
pip freeze > requirements.txt
```

Step 2 — Create Basic Flask App

Create **app.py**

```python
from flask import Flask

app = Flask(__name__)

app.config['SECRET_KEY'] = 'supersecretkey'

@app.route("/")
def home():
    return "Finance Tracker Running!"
```

```
if __name__ == "__main__":
    app.run(debug=True)
```

Run it:

```
python app.py
```

Open:

```
http://127.0.0.1:5000
```

You should see:

```
Finance Tracker Running!
```

Step 3 — Database Setup

Create **config.py**

```python
class Config:
    SECRET_KEY = "secretkey"

    SQLALCHEMY_DATABASE_URI = "mysql+pymysql://root:password@localhost/finance_db"
    SQLALCHEMY_TRACK_MODIFICATIONS = False
```

Create database in MySQL

```sql
CREATE DATABASE finance_db;
```

Step 4 — Models

Create **models.py**

```python
```

```python
from flask_sqlalchemy import SQLAlchemy
from flask_login import UserMixin

db = SQLAlchemy()

class User(UserMixin, db.Model):

    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(50), unique=True)
    email = db.Column(db.String(100), unique=True)
    password = db.Column(db.String(200))

class Transaction(db.Model):

    id = db.Column(db.Integer, primary_key=True)
    user_id = db.Column(db.Integer, db.ForeignKey('user.id'))

    amount = db.Column(db.Float)
    category = db.Column(db.String(50))
    type = db.Column(db.String(10))
    date = db.Column(db.Date)
```

Step 5 — AI Insight Engine

Example logic:

Create **ai/insight_engine.py**

Python

```python
def generate_insights(expenses):

    total = sum(expenses.values())

    insights = []

    for category, amount in expenses.items():

        percentage = (amount / total) * 100

        if percentage > 40:
            insights.append(
                f"You are spending {percentage:.1f}% on {category}. Consider reducing it."
            )

    return insights
```

Step 6 — Dashboard

Dashboard will show:

- total income
- total expenses
- category breakdown
- AI suggestions

Example insight:

```
Food: ₹4500
Rent: ₹8000
Travel: ₹2000
```

AI:

```
You spend 45% on rent.
Your food spending is above average.
```

Step 7 — Deployment

Options:

- **Render**
- **Railway**
- **AWS**

Final Resume Project Line

Use this:

AI-Powered Personal Finance Tracker
Built a full-stack personal finance management platform using Flask, MySQL, and SQLAlchemy with secure user authentication, expense tracking, and AI-driven spending insights that analyze transaction data and generate personalized financial recommendations.

If you want, I can also show you

- The **FULL backend code (500+ lines)**
- **Complete database schema**
- **Dashboard with charts**
- **OpenAI integration**
- **How to deploy it so interviewers can open your website**

Just say **"build the full project step by step"** and we'll make this a **resume-level project**.